

Lead Data Scientist: Case Study Task

This case study is designed to assess your technical skills, business acumen, and ability to build an end-to-end data solution. Your task is to develop a system that identifies potential investment targets by combining news sentiment with key financial metrics, and then contextualizes their value against relevant competitors.

Case Study: Building a Sentiment-Driven Investment Scoring Model

Objective

Develop an automated system that uses news sentiment to identify promising investment opportunities, validates them with financial data, and produces a final, unified score. The analysis will conclude by comparing each target company against its peers to provide a clear, explainable investment thesis.

Context

This project mimics a real-world investment workflow:

1. **Market narrative first:** Start with a broad, sentiment-driven search for compelling stories in the market.
 2. **Quant validation:** Ground the stories in financial reality with rigorous quantitative analysis.
 3. **Competitive lens:** Place each opportunity in its competitive context for a defensible assessment.
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Methodology Outline

Part 1: Market Sentiment & Opportunity Identification

Broad Data Ingestion

Aggregate news articles, press releases, and financial headlines from sources like NewsAPI, GNews, or RSS.

```
from gnews import GNews

google_news = GNews()
# Fetch news related to a specific topic, e.g., 'renewable energy'
news = google_news.get_news('renewable energy')
# news[0] will contain the first article's details
```

Natural Language Processing (NLP)

- Apply sentiment analysis at scale to score all collected articles.
- Implement Named-Entity Recognition (NER) to identify companies mentioned.

```

import spacy
from textblob import TextBlob

nlp = spacy.load("en_core_web_sm")
text = "Apple's new iPhone launch was a massive success, according to analysts."

# NER to find organizations
doc = nlp(text)
for ent in doc.ents:
    if ent.label_ == 'ORG':
        print(f"Company Found: {ent.text}")

# Sentiment Analysis
sentiment = TextBlob(text).sentiment
print(f"Sentiment Polarity: {sentiment.polarity}") # -1 (neg) to 1 (pos)

```

Signal Generation

Identify a shortlist of **five companies** with interesting sentiment patterns (e.g., sharp positive inflection, sustained positivity, or negative → positive turnarounds). This becomes your pool of potential opportunities.

Part 2: Quantitative Validation

Targeted Data Collection

For the five shortlisted companies, collect public financial data using `yfinance` or similar.

```

import yfinance as yf

ticker = yf.Ticker("MSFT")

# Explore this object for relevant metrics
company_info = ticker.info

# Example metrics (not exhaustive):
# Valuation: 'trailingPE', 'forwardPE', 'priceToBook'
# Profitability: 'profitMargins', 'returnOnEquity'
# Growth: 'revenueGrowth', 'earningsQuarterlyGrowth'

print(f"Profit Margin for {ticker.ticker}: {company_info.get('profitMargins')}")
print(f"Revenue Growth for {ticker.ticker}: {company_info.get('revenueGrowth')}")

```

Part 3: Integrated Scoring Model

Hybrid Score Development

Design a reproducible, justifiable score that combines sentiment signals with multiple quantitative metrics.

Scoring Execution

Compute the final score for each of the five target companies, making the weighting of sentiment vs. financial metrics explicit. Candidates should clearly explain and justify the chosen weights (e.g., 40% sentiment, 60% fundamentals) and demonstrate how these affect the overall score.

Part 4: Comparable Analysis & Explainability

Peer Identification

For each of the five companies, identify **three relevant public comps**.

Comparative Scoring & Explanation

Run the model on the 15 comparables (3 per target). For each target, present a final analysis that:

- Compares its score to its comps; and
- Explains differences with specific references to sentiment trends and quantitative metrics.

Example: "Company A scored higher than its peers due to strong sentiment around its recent product launch, combined with a P/E ratio that is 15% below the peer average."

Deliverables

- **Technical implementation:** Python script or Jupyter Notebook (`.ipynb`) implementing the full end-to-end pipeline (ingestion → NLP sentiment/NER → shortlist generation → financial validation → integrated scoring → comps analysis).
- **Presentation (5 slides):** Outlining:
 - Problem, objective & data sources
 - Pipeline & model architecture
 - Scoring logic (with explicit weights & rationale)
 - Results: five targets vs. their three comps each (tables/charts)
 - Takeaways & defensible investment theses.
- **Timeline:** Submit within **three (3) days** of receiving this task.